VISUAL INTELLIGENCE

Comparative Analysis of CNN and ScatNet

for Image Classification

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Course: Visual Intelligence

Lung Cancer Histopathological Image Classification

Binary Classification: Benign vs. Adenocarcinoma

1. Introduction

This project presents a comparative study of two image classification approaches applied to lung cancer histopathological images: a Convolutional Neural Network (CNN) and a Scattering Network (ScatNet). The task is binary classification between Benign and Adenocarcinoma tissue samples. Both models were trained and evaluated on the same dataset using 5-fold cross-validation, and their decision-making processes were analysed using Explainable AI (XAI) methods.

CNNs are the dominant paradigm in image recognition, learning hierarchical features directly from data through learned convolution filters. ScatNets, by contrast, use fixed mathematical wavelet transforms (Morlet wavelets) to extract stable, multi-scale representations without requiring learned filters in the feature extraction stage. This comparison investigates whether hand-crafted mathematical features can compete with learned features for medical image classification.

2. Dataset

The dataset consists of 10,000 histopathological lung tissue images sourced from Kaggle. Images were converted to grayscale and resized to 64×64 pixels. The dataset is perfectly balanced with 5,000 images per class.

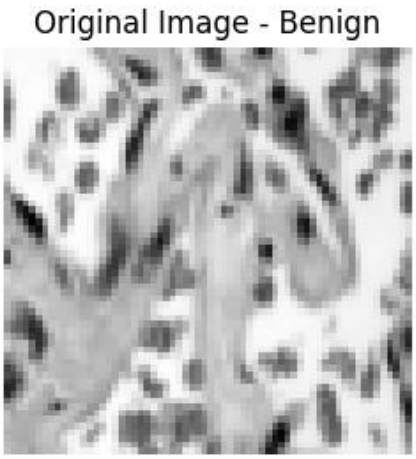

Figure 1: Benign tissue sample

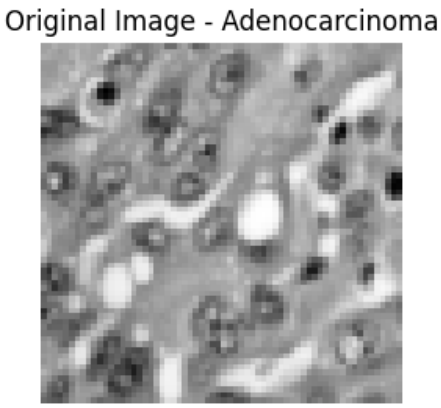

Figure 2: Adenocarcinoma tissue sample

Split	Total Images	Per Class
Training set	8,000	4,000
Test set (held-out)	2,000	1,000
Total	10,000	5,000

Table 1: Dataset split statistics

3. Convolutional Neural Network (CNN)

3.1 Architecture

The CNN (ImprovedCNN) uses three convolutional blocks, each consisting of a Conv2d layer, BatchNorm2d, ReLU activation, and MaxPool2d downsampling. Dropout regularisation (p=0.25) is applied after the convolutional stack, followed by a fully connected classifier with two hidden layers (128 and 64 units) and a 2-class output. Input: 1-channel 64×64 grayscale images.

Layer	Output Channels	Kernel
Conv1 + BN + ReLU + Pool	16	3×3
Conv2 + BN + ReLU + Pool	32	3×3
Conv3 + BN + ReLU + Pool	64	3×3
Dropout (p=0.25)	—	—
FC Classifier	128 → 64 → 2	—

Table 2: CNN architecture summary

3.2 Learned Filters

The 16 filters learned by the first convolutional layer capture a variety of low-level features including edges, textures, and frequency patterns across different orientations.

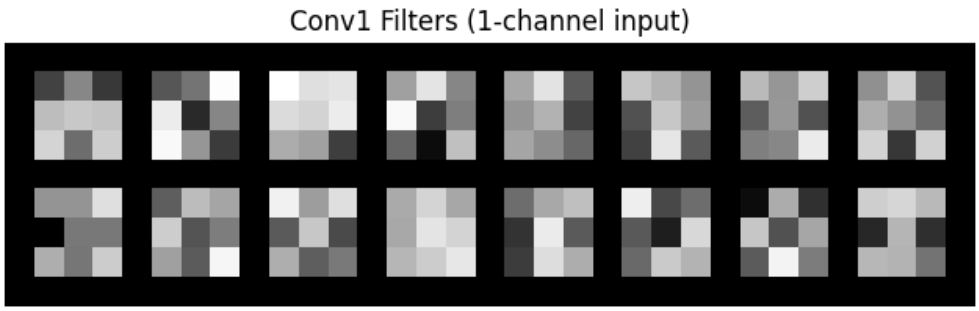

Figure 3: Learned Conv1 filters (16 filters, 1-channel input)

3.3 Training Results

Training used 5-fold cross-validation with 10 epochs per fold, Adam optimiser (lr=0.001, weight decay=1e-4), batch size 16, and CrossEntropyLoss. The best model across folds achieved 99.56% validation accuracy. Overall test accuracy on the held-out set is 99.00% — 983/1,000 Benign and 995/1,000 Adenocarcinoma correct.

Metric	Benign	Adenocarcinoma
Precision	0.99	0.98
Recall	0.98	0.99
F1-Score	0.99	0.99

Table 3: CNN classification report — overall test accuracy: 99.00%

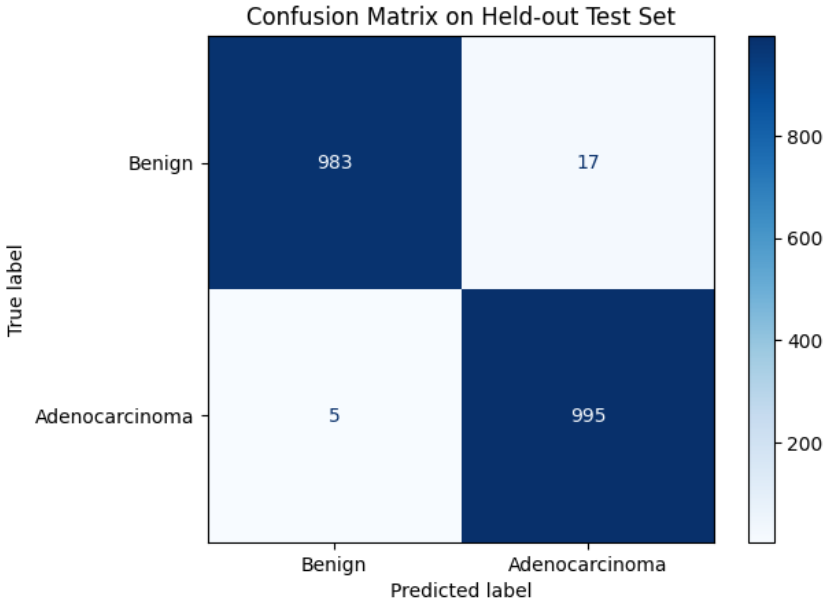

Figure 4: Confusion matrix — CNN on held-out test set

4. Scattering Network (ScatNet)

4.1 Architecture and Feature Extraction

The ScatNet uses the Kymatio library's Scattering2D transform with J=2 scale levels and L=8 orientations, applied to 64×64 grayscale images. The transform produces 81 scattering coefficients per image, organised by order: 1 zeroth-order coefficient (global average energy), 16 first-order coefficients (J×L = 2×8, edges at different scales and orientations), and 64 second-order coefficients (texture interactions). Each coefficient is a 16×16 spatial map, globally mean-pooled to an 81-dimensional feature vector.

This feature vector is fed into a small classifier (ScatNetClassifier) with two fully connected hidden layers (128 and 64 units) and a 2-class output, trained with Dropout (p=0.5).

4.2 Morlet Wavelet Filters

Unlike the CNN, the ScatNet uses fixed Morlet wavelet filters that are not learned from data. The filters are shown below for both scale levels (j=0,1) across all 8 orientations (θ=0 to 7). Colour saturation and hue encode the magnitude and phase of each complex filter.

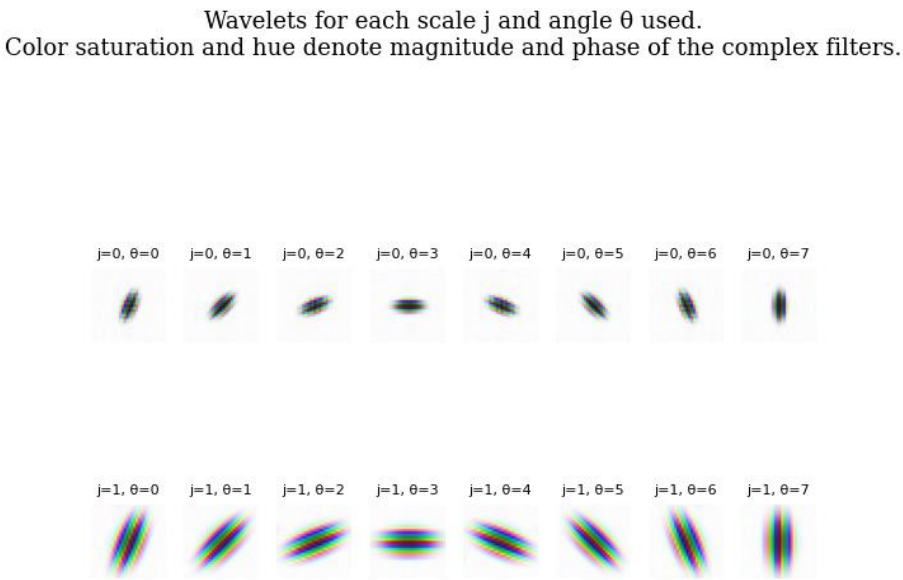

Figure 5: Morlet wavelet filters at J=2 scales and L=8 orientations

4.3 Scattering Feature Maps

The 81 scattering coefficient maps produced for a single input image are shown below. The first map (top-left) is the zeroth-order coefficient representing the smoothed image energy. Subsequent maps capture responses at progressively finer orientations and scales.

Figure 6: All 81 scattering feature maps (9×9 grid) for one test image

4.4 Training Results

The ScatNet classifier was trained using the same 5-fold cross-validation protocol with 10 epochs per fold, Adam optimiser (lr=0.001), and batch size 64. Mean accuracy across folds was 88.31% and mean F1 score was 88.28%.

Metric	Value
Mean Val Accuracy (5 folds)	88.31%
Mean Val F1 Score (5 folds)	88.28%
Test Accuracy	88.60%
Test F1 Score (macro)	0.88
Trainable Parameters	~22K

Table 4: ScatNet performance on held-out test set

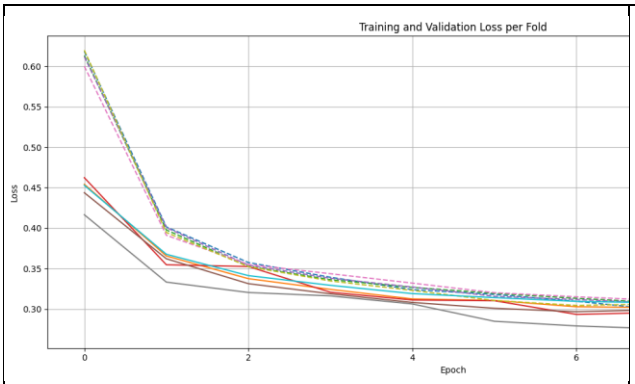

Figure 7: Training and validation loss across 5 folds

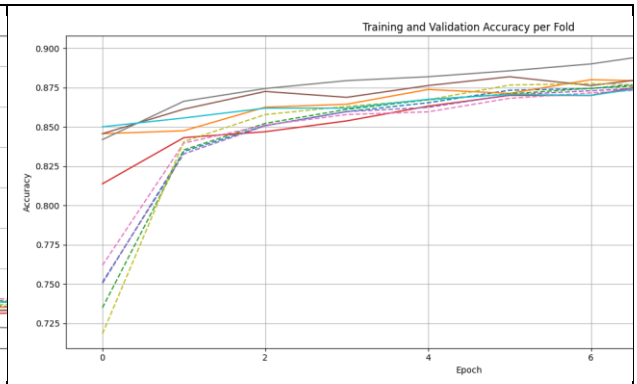

Figure 8: Training and validation accuracy across 5 folds

The training curves show stable convergence across all 5 folds. The validation loss decreases consistently with no evidence of severe overfitting. However, ScatNet plateaus at around 88% accuracy, well below the CNN's 99%, suggesting the fixed scattering features do not capture all discriminative information present in the histopathological images.

5. Explainable AI (XAI)

5.1 Method: Patch Permutation

Patch permutation importance measures how much each spatial region of the input image contributes to the model's prediction. The image is divided into 16×16 patches; for each patch the pixel values are randomly shuffled and the drop in predicted probability for the correct class is recorded. This is repeated 5 times per patch and averaged. Higher values (brighter in the heatmap) indicate regions the model relies on most. This method was applied to both CNN and ScatNet on 10 images per class.

5.2 Heatmaps — Benign Class

The CNN attribution (centre) shows high importance in specific localised patches corresponding to structural tissue features. The ScatNet attribution (right) shows a more distributed pattern, reflecting that global scattering coefficients integrate information differently.

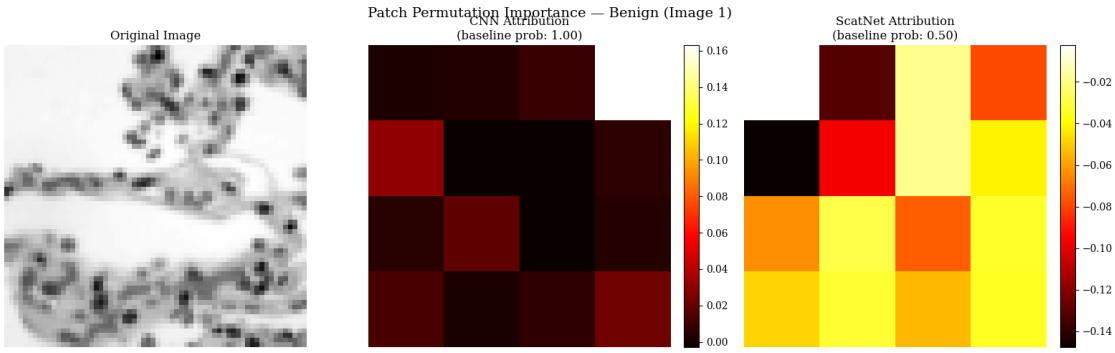

Figure 9: Patch permutation heatmap — Benign (Image 1). Left: original. Centre: CNN attribution. Right: ScatNet attribution.

5.3 Heatmaps — Adenocarcinoma Class

For Adenocarcinoma images, the CNN focuses strongly on characteristic dense circular glandular structures visible in the tissue. The ScatNet attribution distributes importance more uniformly, consistent with its global pooling of scattering coefficients.

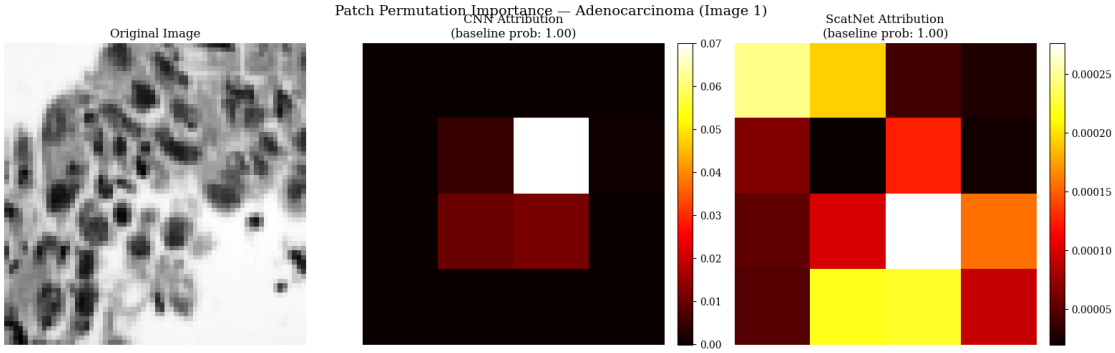

Figure 10: Patch permutation heatmap — Adenocarcinoma (Image 1). Left: original. Centre: CNN attribution. Right: ScatNet attribution.

6. ScatNet Feature Importance

6.1 Method: Captum FeaturePermutation

For the ScatNet, feature permutation importance was computed using Captum's FeaturePermutation method. Each of the 81 scattering coefficients is individually shuffled across the batch and the resulting change in output score measured. Attribution was computed separately for each class (1,000 samples) and averaged to produce global importance scores.

6.2 Importance Across All 81 Coefficients

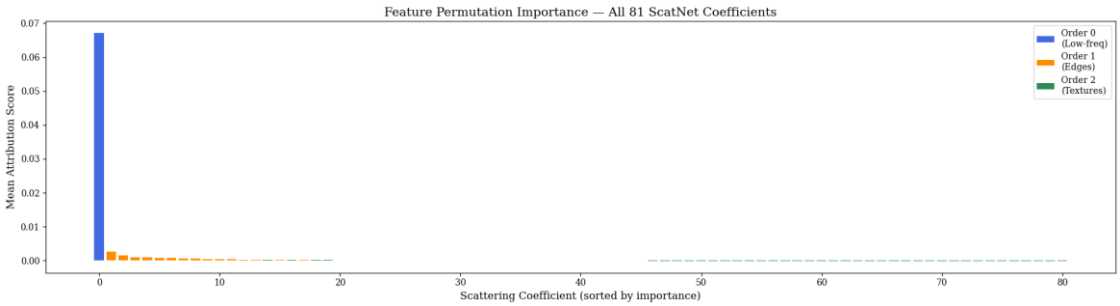

Figure 11: Feature permutation importance for all 81 scattering coefficients (sorted by importance, colour-coded by order)

The results show a striking pattern: the single zeroth-order coefficient (Order 0) dominates with an importance score approximately 10× higher than any first-order coefficient. The global average energy of the image is by far the most predictive scattering feature for distinguishing Benign from Adenocarcinoma tissue.

6.3 Importance by Scattering Order

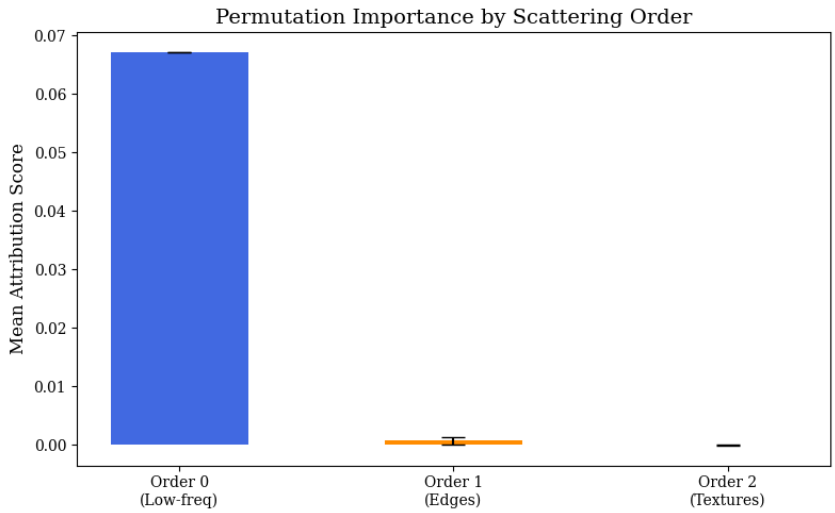

Figure 12: Importance grouped by scattering order

Scattering Order	Description	Mean Importance
Order 0	Global low-frequency energy (1 coeff)	~0.067
Order 1	Edges and orientations (16 coeffs)	~0.001
Order 2	Texture interactions (64 coeffs)	~0.000

Table 5: Mean permutation importance by scattering order

6.4 Per-Class Comparison

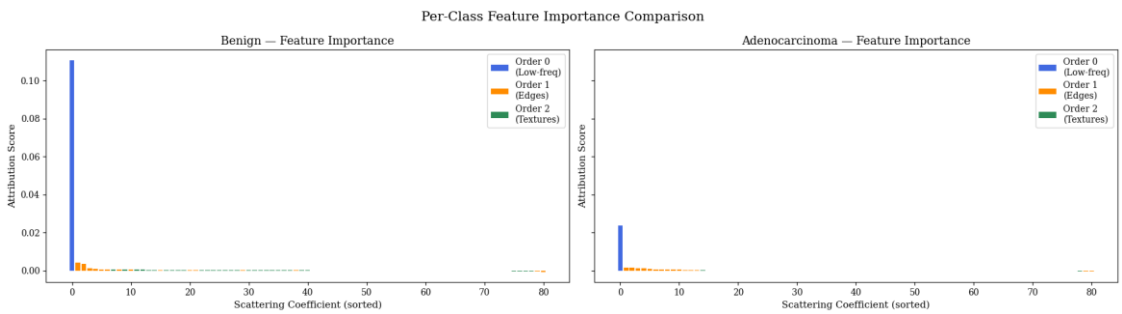

Figure 13: Feature importance comparison between Benign (left) and Adenocarcinoma (right) classes

The per-class breakdown shows that Order 0 dominance is consistent across both classes. The overall energy level of the image is the primary discriminative feature for the ScatNet regardless of the class being predicted.

7. Model Comparison and Discussion

7.1 Performance Summary

Metric	CNN	ScatNet
Test Accuracy	99.00%	88.60%
F1 Score (macro)	0.99	0.88
Precision (macro)	0.99	0.88
Recall (macro)	0.99	0.89
False Positives (test)	17	~138
False Negatives (test)	5	~92

Table 6: Performance comparison on held-out test set (2,000 images)

7.2 Discussion

The CNN outperforms the ScatNet by a significant margin (99% vs 88.6% accuracy). This is expected given that CNNs learn task-specific features directly from data, while ScatNet uses fixed mathematical transforms. However, the ScatNet achieves this result with a far smaller classifier (~22K trainable parameters vs ~660K for the CNN), making it computationally efficient.

The XAI analysis reveals fundamentally different decision strategies. The CNN uses spatially localised regions of the image to make predictions, focusing on specific tissue structures. The ScatNet, however, relies overwhelmingly on the global energy level (zeroth-order coefficient), with minimal contribution from edge or texture features — essentially performing a coarse global intensity discrimination rather than analysing fine structural patterns.

This finding explains the performance gap: Adenocarcinoma tissue may have systematically different global intensity distributions from Benign tissue, which the ScatNet can detect. But the CNN additionally exploits local structural differences (glandular morphology, nuclear density) that the scattering coefficients, particularly after global mean pooling, fail to preserve.

7.3 Conclusion

This project demonstrates that while ScatNets provide mathematically principled and interpretable feature representations, CNNs are significantly more powerful for fine-grained medical image classification tasks. The ScatNet's reliance on low-frequency global features limits its ability to exploit the rich local structural information present in histopathological images. Future work could explore combining scattering features with learned features, or using spatial scattering maps (without mean pooling) to preserve local information.

References

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